

Ostbayerische Technische Hochschule Amberg-Weiden
Fakultät Elektrotechnik, Medien und Informatik

Studiengang Bachelor Künstliche Intelligenz

Bachelorarbeit

von

Sebastian Weidner

**Imitation Learning unter Verwendung
videokonditionierter Policies für visuell gesteuertes
Roboterverhalten**

Imitation Learning using Video-Conditioned Policies for
Visually Guided Robotic Behavior

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Studiengang: **Bachelor Künstliche Intelligenz**

Ich bestätige, dass ich die Bachelorarbeit mit dem Titel:

**Imitation Learning unter Verwendung videokonditionierter Policies für visuell
gesteuertes Roboterverhalten**

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Bachelorarbeit Zusammenfassung

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Ausgabedatum: 20. Januar 2026	Abgabedatum: 20. Juni 2026

Titel:

Imitation Learning unter Verwendung videokonditionierter Policies für visuell gesteuertes Roboterverhalten

Zusammenfassung:

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Schlüsselwörter: $\text{\LaTeX}$, Textsatz, Formeln, Graphiken

Abstract:

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Keywords: $\text{\LaTeX}$, Textsatz, Formeln, Graphiken

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Acronyms

IL Imitation Learning

LfV Learning from Videos

LLM Large Language Model

RL Reinforcement Learning

VLA Vision-Language-Action

Mathematical Notation in Formulas

Capital Letters

Capital Letters represent *random variables*.

Examples:

- S : Represents the set of all states in which the robot can be.
- A : Represents the set of all possible actions the robot can perform.

Lowercase Letters

Lowercase letters represent *specific values* of *random variables*.

Examples:

- $s \in S$: Stands for a specific robot state.
- $a \in A$: Stands for a specific action that the robot performs.

Timestep t . A lowercase letter is usually followed by a time index t , representing the specific action taken by the robot at time step t . Examples:

- $s_t \in S$: Stands for a specific robot state at time step t .
- $a_t \in A$: Stands for a specific action that the robot performs at time step t .
- In some cases the time index is omitted for simplicity, e.g. a instead of a_t .
- In the literature, the next state or action is sometimes denoted as s' or a' instead of s_{t+1} or a_{t+1} .

Capital Letters in Bold

Capital letters in bold represent matrices.

For example:

$$\mathbf{W} = \begin{bmatrix} w_{11} & w_{12} & \dots & w_{1n} \\ w_{21} & w_{22} & \dots & w_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ w_{m1} & w_{m2} & \dots & w_{mn} \end{bmatrix} \quad (1)$$

Lowercase Letters in Bold

Lowercase letters in bold represent vectors.

For example:

$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} \quad (2)$$

Chapter 1

Experiments on a Real Robot

The second part of this work will focus on the practical implementation of an open-source Imitation Learning (IL) approach on a real robotic system. The goal is to gain hands-on experience with the challenges and opportunities how to run a robot model on a real robot. This includes finding out how to connect a robot with the PC, how to control the robot from the PC and what additional code is needed to adapt a robot model to run on our specific robot. The idea was to implement a easy-to use basic pipeline to communicate with the robot, with an framework where only the robot model must be replaced. This includes also a camera setup with camera calibration to capture the robot's environment to provide the necessary visual input for the robot model. The whole pipeline should be designed to be modular and extensible, allowing for easy integration of different robot models with different software dependencies.

Robot Hardware

Robot: UR5e Robot Arm from Universal Robots Purpose Collaborative robot (cobot) designed for safe human-robot interaction

Robot: UR5e Robot Arm from Universal Robots	
Degrees of freedom	6 rotating joints
Payload	5 kg
Reach	850 mm
Weight	12 kg
Materials	Powder coated steel

Table 1.1: Specifications of the UR5e Robot Arm

- Degrees of freedom: 6 rotating joints
- Programming: 12 inch touchscreen with PolyScope Graphical user interface

- Payload: 5 kg
- Reach: 850 mm
- Collaborative robot (cobot) designed for safe human-robot interaction
- Weight: 12 kg
- Materials: Powder coated steel

- Freiheitsgrade: 6 degrees of freedom - Payload capacity of 5 kg - Reach of 850 mm - Collaborative robot (cobot) designed for safe human-robot interaction

The implementation will be based on the UR5e robot arm, which is a widely used robotic platform in research and industry. The UR5e provides a versatile and user-friendly interface for controlling the robot, making it an ideal choice for implementing and testing IL methods.

implement the communication. This will provide valuable insights into the practical aspects

of deploying a robot IL methods in real-world settings and to evaluate the performance of different approaches on a physical robot. The UR5e robot arm will be used for this purpose, and the implementation will involve connecting the

Connect Robot with PC

ROS2 Connection

RTDE connection

Chapter 2

Summary, Conclusion & Outlook

2.1 Summary

This thesis examined imitation learning as a promising approach for enabling robots to acquire complex skills from demonstrations and move toward the long-term goal of general-purpose robotic intelligence. Large Language Models (LLMs) have demonstrated that training powerful architectures on vast and diverse datasets can lead to remarkable generalization capabilities, enabling systems to solve tasks for which they were never explicitly trained. A central goal of imitation learning is to bring similar capabilities to robotic systems by leveraging the enormous amount of video data available on the internet.

Chapter ?? introduced the fundamental concepts of imitation learning and distinguished between two major paradigms: Behavior Cloning and Learning from Videos (LfV). While Behavior Cloning relies on demonstrations containing explicit action labels, Learning from Videos seeks to exploit the vast amount of action-free video data available online, offering a scalable alternative for robot learning.

Chapter ?? discussed the overarching objective of robotics research: the development of generalist robots capable of performing a wide range of tasks in unstructured and previously unseen real-world environments. A major obstacle to this vision is the robotics data bottleneck, which arises from the limited availability of large, diverse, and high-quality robotic datasets. Internet videos represent a potential solution due to their abundance and diversity; however, their use introduces additional challenges related to the transfer of visual knowledge to robotic systems.

A prerequisite for effective human-robot interaction is the ability to communicate goals and intentions to robotic agents. Therefore, chapter ?? examined how robots can be instructed and guided to perform desired tasks.

Chapter ?? focused on the central research questions of Learning from Videos: how relevant knowledge can be extracted from internet videos and how this knowledge can subsequently be transferred to robotic systems for policy learning and control. Video Foundation Models were introduced as a powerful tool for extracting semantic

and behavioral information from large-scale video data. Two major challenges were identified in the transfer process: the absence of action labels and the distribution shift between human videos and robotic observations. To address these challenges, alternative action representations are employed to relabel action-free video data, enabling robots to learn from demonstrations that do not contain explicit control signals.

Chapter ?? explored the role of robust video understanding in IL. Learning manipulation skills from video requires more than simple action recognition: (1) It demands meaningful representations of the environment, (2) an understanding of object affordances and human-object interactions, (3) recognition of human activities, and (4) accurate modeling of fine-grained hand movements. These capabilities form the basis for a general-purpose robot that can interpret human demonstrations.

Chapter ?? focused on approaches for learning coherent robot models from video data. (1) The chapter first introduced foundational perception and representation learning methods and techniques. (2) It then discussed both RL-based and imitation learning approaches, highlighting their respective strengths and limitations. (3) Hybrid methods, which combine the structured guidance and sample efficiency of imitation learning with the exploration and optimization capabilities of reinforcement learning. Special attention was given to the intersection of Learning from Videos (LfV) and Reinforcement Learning (RL), demonstrating how knowledge extracted from videos can be used to improve policy learning and enhance the sample efficiency of RL algorithms. (4) Last but not least, the chapter presented multi-modal learning approaches and Vision-Language-Action (VLA) models and the three Architectural paradigms, including sensorimotor, world-model-based, and affordance-based architectures. VLA models represent currently the state of the art in video-based robot learning.

Chapter ?? examined the major challenges and future research directions in IL and video-based robotic learning. Despite significant advances in recent years, the development of robust and generalizable robotic systems remains constrained by several fundamental limitations, including data availability, computational scalability, domain transfer, and efficient policy learning. These challenges highlight the gap between current research systems and the long-term goal of general-purpose robotic intelligence. At the same time, emerging approaches such as diffusion-based policies, world models, and causally-aware learning systems offer promising opportunities for overcoming these limitations and improving the adaptability, reasoning capabilities, and generalization performance of future robotic agents.

Finally in Chapter ??, selected IL approaches deployed on real robotic systems were examined to provide practical insight into the theoretical concepts discussed throughout the thesis. These examples illustrated how IL can enable robots to acquire complex manipulation and control skills directly from video demonstrations.

2.2 Outlook

Implementing and testing a open-source Robot model on our UR5e Robot will be the next step. This will provide practical insights into the challenges of deploying IL approaches on real robotic systems and allow for the evaluation of different methods in a real-world setting. Implementing the communication to the robot with RTDE will be the first step, then it can be done with ROS2.

There exists many different Architectures which could be used: Timesformer, Q-Former, Use of Spike Neural Networks, Graph Neural Networks and Causal Reasoning. There exist many different Architectures of VLA and Multi-Modal Learning / Models. Action Chunking Transformer which is maybe better than BC: ACT is preferred when you want a robust, smooth, long-horizon policy distilled from demonstrations, whereas standard BC may fail if the task requires multi-step reasoning. - Conditional Behavior Transformers (C-BeT) - Forschungsrichtung: Affordance grounding for robot interaction - conditional variational autoencoder (C-VAE)

Other Research directions for next works are: The field of Robotics provides a large field of research. Further work could examine the available Datasets and categories. Examine more implemented successful Architectures to become a better understanding, what is possible. There is a lot of Research of VLA Models to robotics. Meta-Learning Approaches could be examined. The integration of Causal Reasoning and World Models could be further explored.

RL techniques and how they can combined with IL approaches, for example through finetuning to increase task success.

There is no work comprehensively comparing the utility of different action representations for LfV.

LfV Policy Methods: - **Future Directions:** - Jointly pretraining **multi-modal foundation models** (on both video and robot data) remains underexplored. - Use **spatio-temporal encoders** and large internet datasets. - representations from video foundation models Methods that leverage representations $\hat{A}$ will be limited by the scalability of their $\hat{A}$ (discussed in Section 4.2.1). An open question is whether $\hat{A}$'s are best utilised explicitly in a hierarchical setup, or if they are most useful for representation transfer. In hierarchical setups, where the policy is defined as $\pi(a_t | \pi_{alt}(\hat{a}_t | o_t), o_t)$, we should explore the extent to which either π_{alt} or π is the performance bottleneck.

Dynamics Models: scalable alternative-action techniques will be useful for conditioning foundational video predictors. - Exploring **hierarchical world/dynamics models** (higher levels learned from video, lower levels from robot data) is a promising avenue - An underexplored direction is to combine standard analytic simulation with video prediction simulation

- **Category: Visual Question Answering (VQA):** - Ask video-to-text models questions like "Did the robot complete the task?" $\rightarrow$ reward = task completion or score progress (Dense Reward) - VQA only used with images as input, but not with video-based VQA Following synthetic data trends in other fields, future work may seek to use

foundational video predictors to generate synthetic video rollouts, or augment and improve the coverage of existing data via video editing

Reward Functions: Underexplored direction: is to use LfV rewards to augment reward labels in offline RL data.

- **Safety:** LfV reward may be useful for detecting **safety-related metrics** - **RL-AI-F:** Extracting shaped reward functions from video-to-text models via **RL from AI feedback (RL-AI-F) methods** is an avenue for future work.

- **Pretrained Foundation Models:** The most promising approaches - including **video-language models** and **video predictors** as reward functions - **Integration:** LfV reward functions could be combined with other LfV KMs,

Value Functions for example, by finetuning an LfV policy through online RL using LfV rewards. TD-learning of value functions from large-scale human video is seen as a promising direction for real-world robotics

Scarce research: few works on value learning from video

Monolithic vs Compositional Models —> compare Approaches - video generation model to hierarchically condition an action-prediction —> example compositional LfV approach - Any-to-any sequence modelling —> example monolithic LfV approach - compositional approaches are more computationally efficient, more data-efficient, and can obtain improved generalization - monolithic models may benefit from positive transfer between the different data modalities and tasks they handle, and can be optimized end-to-end.

Recovering action information from video

1. Explicitly compare the pros and cons of different action representations, and measure the extent to which each can help obtain the benefits of LfV (see Section 3.3).
2. Develop improved, novel action representations. Perhaps by combining the benefits of existing options [310].

Benchmark - Improved benchmarks that quantitatively measure LfV benefits (Section 5.2.2) will facilitate these efforts

We need metrics that capture not only task success but also generalization, robustness, and sim-to-real performance.

- We consider task goals to be described by contact states. - Dadurch werden auf natürliche Weise Mehrdeutigkeiten vermieden, die bei der Betrachtung räumlicher Beziehungen auftreten - **Das Erschließen menschlicher Absichten bei gleichzeitiger Beseitigung der inhärenten Mehrdeutigkeiten in Videos ist eine zukünftige Richtung, die es zu erforschen gilt.** - We have also assumed that demonstration videos must be captured with a stationary camera. In reality, most videos in everyday scenarios contain a moving camera. - Daher besteht eine vielversprechende zukünftige Richtung darin, zu untersuchen, wie sich ein Modell erstellen lässt, das dynamische Szenen aus einer bewegten Kamera rekonstruieren kann, wobei das gewünschte Modell die Geometrie sowohl statischer als auch bewegter Objekte in den Szenen schätzen kann

und gleichzeitig sicherstellt, dass der Maßstab der rekonstruierten Szenen und Objekte der realen Welt entspricht.

Chapter 3

Imitation Learning fundamentals

Was sollte noch erklärt werden?

Online Learning / Offline Learning

Multi-Modal und Foundation Models

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